Slide Set #2

Tools

Pietro Coretto
pcoretto@unisa.it

Econometrics
Master in Economics and Finance (MEF)
Università degli Studi di Napoli "Federico II"

Version: Tuesday 5th January, 2021 (h11:30)

Notes

Vector spaces

A vector space (linear space) is a collection of objects, called vectors, that can be added together, or multiplied by a scalar (scaling). These operations need to satisfy certain axioms.

Many objects in mathematics can be organized in vector spaces, and this has the advantage that we can define concepts like *distance*, *length/size* or *angles* in a general way.

The latter is useful to extend the notion of *parallelism*, *direction*, *magnitude*, etc., to a broad class of objects.

By now we focus on the Euclidean space $\mathbb{R}^K$, $\boldsymbol{x} = (x_1, x_2, \ldots, x_K)'$ is a K-dimensional vector of $\mathbb{R}^K$.

Notes

Distances and metrics

A *metric space* is a space where we can measure distance between its objects.

A **distance** is an abstract function $d(\cdot,\cdot)$ that measures of how far apart objects. A distance needs to fulfill axioms (positive-definiteness, symmetry, triangle inequality).

The Euclidean vector space $\mathbb{R}^K$ is also a metric space, examples of distances are

- $d_1(\boldsymbol{x},\boldsymbol{y}) = \sum_{i=1}^K |x_i - y_i|$
- $d_2(\boldsymbol{x},\boldsymbol{y}) = \sqrt{\sum_{i=1}^K (x_i - y_i)^2}$
- $d_\infty(\boldsymbol{x},\boldsymbol{y}) = \max\{|x_i - y_i|, i = 1, 2, \ldots, K\}$
- $d_p(\boldsymbol{x},\boldsymbol{y}) = \left(\sum_{i=1}^K |x_i - y_i|^p\right)^{\frac{1}{p}}$, where $p \geq 1$

Notes

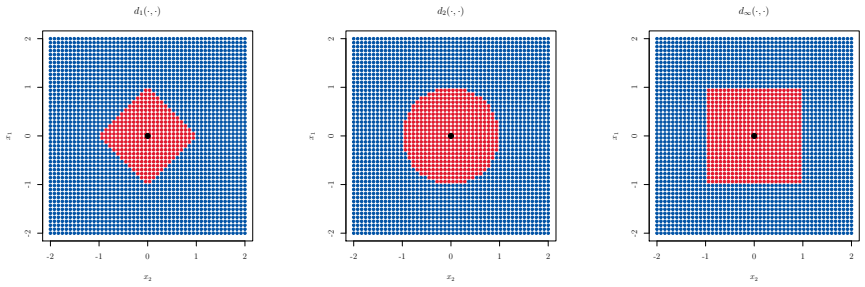
Every distance defines a different way of *metricizing* the space

Take $\mathbb{R}^2$, and consider the point $\mathbf{0} = (0,0)'$. Fix $\varepsilon > 0$, and think about "open balls" (neighbourood) of the point $\mathbf{0}$

$$N_1 = \left\{\boldsymbol{x} \in \mathbb{R}^2 : d_1(\boldsymbol{x},\mathbf{0}) \leq \varepsilon\right\}$$
$$N_2 = \left\{\boldsymbol{x} \in \mathbb{R}^2 : d_2(\boldsymbol{x},\mathbf{0}) \leq \varepsilon\right\}$$
$$N_\infty = \left\{\boldsymbol{x} \in \mathbb{R}^2 : d_\infty(\boldsymbol{x},\mathbf{0})_\infty \leq \varepsilon\right\}$$

Notes

Take points $\boldsymbol{x} = (x_1, x_2)' \in [-2, 2] \times [-2, 2]$, fix $\varepsilon = 1$, and color in red those points in the open balls



Notes

Length, size, magnitude

A **norm** of a vector it's a non negative function $\|\cdot\|$ that measures the "*length/size/magnitude*" of a vector. The norm needs to fulfill a set of axioms.

Vector space with a norm, are called *normed vector space*.

Various norms for the usual Euclidean space

- ℓ_1-norm: $\|\boldsymbol{x}\|_1 = \sum_{i=1}^K |x_i|$
- ℓ_2-norm: $\|\boldsymbol{x}\|_2 = \sqrt{\sum_{i=1}^K x_i^2}$
- ℓ_∞-norm: $\|\boldsymbol{x}\|_\infty = \max\{|x_i|, i = 1, 2, \ldots, K\}$
- ℓ_p-norm: $\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^K |x_i|^p\right)^{\frac{1}{p}}$, where $p \geq 1$

Notes

There is an important connection between *metric space* and *normed vector space*

For given norm $\|\cdot\|$ we can define a distance:

$$d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|$$

In the Euclidean $\mathbb{R}^K$ space

- $d_1(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_1 = \sum_{i=1}^K |x_i - y_i|$
- $d_2(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_2 = \sqrt{\sum_{i=1}^K (x_i - y_i)^2}$
- $d_\infty(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_\infty = \max\{|x_i - y_i|, i = 1, 2, \ldots, K\}$
- $d_p(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_p = \left(\sum_{i=1}^K |x_i - y_i|^p\right)^{\frac{1}{p}}$, $p \geq 1$

Also note that this clarifies the meaning of *norm* as measure of *length/size/magnitude*. Let $\mathbf{0}$ be the *origin*, a fixed reference point of the vector space, then

$$\|\boldsymbol{x}\| = \|\boldsymbol{x} - \mathbf{0}\| = d(\boldsymbol{x}, \mathbf{0})$$

Notes

Angles

An **inner product** it's another function $\langle \cdot, \cdot \rangle$ useful for measuring angles between vectors. It also needs to fulfill axioms.

The inner product (aka dot product) defined on the Euclidean space $\mathbb{R}^K$ is

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{x}'\boldsymbol{y} = \sum_{i=1}^K x_i y_i$$

If we have an inner product for a vector space, we also have norm. Take the Eucliden space:

$$\|\boldsymbol{x}\|_2 = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle} = \sqrt{\sum_{i=1}^K x_i^2}$$

Norm + inner product define angles:

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \cos(\theta) \|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2$$

where θ is the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$.

Notes

The *Cauchy–Schwarz Inequality* is a direct consequence of this equation, in fact:

$$\cos(\theta) = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2} \in [-1, 1]$$

Orthogonality: $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal/perpendicular when $\theta = 90°, 270°$, that is $\cos(\theta) = 0$. $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal **if and only if** $\boldsymbol{x}'\boldsymbol{y} = 0$

Collinearity. $\boldsymbol{x}$ and $\boldsymbol{y}$ are collinear if they are either along same line or are parallel to each other. I this case $\theta = 0°, 180°$ which means $\cos(\theta) = 1$ or $\cos(\theta) = -1$.

Notes

Linear dependence

The vectors in a subset $\mathbb{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_k\}$ of a vector space are *linearly dependent* (LD) if there exist scalars $\{a_1, a_2, \ldots, a_k\}$, not all of them null, such that

$$a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \ldots, a_k\boldsymbol{x}_k = \boldsymbol{0}$$

therefore, assuming (wlog) $a_1 \neq 0$,

$$\boldsymbol{x}_1 = \frac{-a_2}{a_1}\boldsymbol{x}_2 + \frac{-a_3}{a_1}\boldsymbol{x}_3 +, \ldots, + \frac{-a_k}{a_1}\boldsymbol{x}_k$$

The set of vectors in $\mathbb{X}$ are *linearly independent* (LI) if the equation

$$a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \ldots, a_k\boldsymbol{x}_k = \boldsymbol{0}$$

can only hold for $a_1 = a_2 =, \ldots, = a_k = 0$

A set $\mathbb{X}$ is LI if it is not redundant, in the sense that you cannot express any vector of $\mathbb{X}$ as a linear combination of the other elements of $\mathbb{X}$.

Notes

A subset $\mathbb{S}$ *spans* a vector space $\mathbb{V}$ if you can build any element of $\mathbb{V}$ as a linear combinations of vectors in $\mathbb{S}$

A subset $\mathbb{S}$ of a vector space $\mathbb{V}$, is a basis if it is LI and it spans $\mathbb{V}$

Example. Take the units vectors in $\mathbb{R}^2$, that is

$$\boldsymbol{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \boldsymbol{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

It is easy to see that $\mathbb{S} = \{\boldsymbol{e}_x, \boldsymbol{e}_y\}$ is **basis** for $\mathbb{R}^2$, in fact, any point $\boldsymbol{p} \in \mathbb{R}^2$ can be written as a linear combination of elements of $\mathbb{S}$. For example

$$\boldsymbol{p} = \begin{pmatrix} a \\ b \end{pmatrix} = a\boldsymbol{e}_x + b\boldsymbol{e}_y$$

for any two real numbers (a, b)

Notes

Orthogonality

Assume $\mathbb{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_K\}$ is a set of non-zero vectors that are orthogonal (pairwise), then

1. vectors of $\mathbb{X}$ are LI
2. $\mathbb{X}$ is a basis for $\mathbb{R}^K$, in particular an orthogonal basis

In the example before $\mathbb{S} = \{\boldsymbol{e}_x, \boldsymbol{e}_y\}$ is an orthogonal basis for $\mathbb{R}^2$

Having an orthogonal basis for a vector space $\mathbb{V}$ is terribly cool! Why?

It means that we can *compress* $\mathbb{V}$ in a convenient way. Namely,

- each $\boldsymbol{x} \in \mathbb{V}$ can be written as a linear combination of the elements of the orthogonal basis
- each $\boldsymbol{x} \in \mathbb{V}$ can be separated in *independent* contributions

Notes

Let's consider (wlog) $\mathbb{R}^2$, and take two basis

$$\mathbb{S}_t = \{\boldsymbol{t}_1, \boldsymbol{t}_2\}, \quad \text{and} \quad \mathbb{S}_u = \{\boldsymbol{u}_1, \boldsymbol{u}_2\}$$

where only $\mathbb{S}_u$ is an *orthogonal basis*.

Now consider any point $\boldsymbol{x} \in \mathbb{R}^2$. Since these are basis then it must be true that there will be scalars (a, b) and (c, d) such that

$$\boldsymbol{x} \;=\; a\boldsymbol{t}_1 + b\boldsymbol{t}_2 \;=\; c\boldsymbol{u}_1 + d\boldsymbol{u}_2$$

Question: given $\boldsymbol{x}$, can we always determine its components (a, b) or (c, d)? Do these components uniquely relate to $\boldsymbol{x}$?

Notes

For the orthogonal basis

$$\langle \boldsymbol{x}, \boldsymbol{u}_1 \rangle = \langle c\boldsymbol{u}_1 + d\boldsymbol{u}_2, \boldsymbol{u}_1 \rangle = c \langle \boldsymbol{u}_1, \boldsymbol{u}_1 \rangle + d \langle \boldsymbol{u}_2, \boldsymbol{u}_1 \rangle = c \langle \boldsymbol{u}_1, \boldsymbol{u}_1 \rangle$$
$$\langle \boldsymbol{x}, \boldsymbol{u}_2 \rangle = \langle c\boldsymbol{u}_1 + d\boldsymbol{u}_2, \boldsymbol{u}_2 \rangle = c \langle \boldsymbol{u}_1, \boldsymbol{u}_2 \rangle + d \langle \boldsymbol{u}_2, \boldsymbol{u}_2 \rangle = d \langle \boldsymbol{u}_2, \boldsymbol{u}_2 \rangle$$

Therefore, given $\boldsymbol{x}$, we can always find

$$c = \frac{\langle \boldsymbol{x}, \boldsymbol{u}_1 \rangle}{\langle \boldsymbol{u}_1, \boldsymbol{u}_1 \rangle} \qquad \text{and} \qquad d = \frac{\langle \boldsymbol{x}, \boldsymbol{u}_2 \rangle}{\langle \boldsymbol{u}_2, \boldsymbol{u}_2 \rangle}$$

However, this doesn't happen for the non-orthogonal basis $\mathbb{S}_t$.

This example extends to more general spaces, not just $\mathbb{R}^K$, ant it teaches us the beauty of the concept:

orthogonality = possibility to represent each member of the space with a linear decomposition where its individual components can be uniquely identified

Notes

Beauty of vector spaces with an inner product

If a vector space is endowed with a norm, then it has a distance

But recall that if a vector space has an inner product, we can use it define a norm... and a distance as a consequence.

Therefore, having an inner product means that we can measure angles, size, distance, and define orthogonality.

Whatever complicated the vector space is, we know that if it has an orthogonal basis, its very simple to map all its elements into simpler individual components!

Notes

Matrix algebra

Algebra: is a set of objects that can be combined with one or more operations based on rules that make the results well defined.

Matrix sum, matrix product, along with the determinant and the inverse allow to construct an algebraic system where one can mimic the fact that with real numbers:

- there exists unitary element such that $1 \times x = x \times 1 = x$ for any $x \in \mathbb{R}$
- one can construct the reciprocal x^* is such that $x \times x* = x * \times x = 1$, in $\mathbb{R}$ you can set $x^* = 1/x$ for any $x \neq 0$

All the burden of certain matrix algebra calculations are essentially needed to achieve these kind of things.

Notes

A $m \times n$ matrix is a 2-dimensional array, we will look at matrices with real numbers.

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij}) \in \mathbb{R}^{m \times n}$$

Matrix can be seen as sets of vectors:

- jth column vector $(a_{1j}, a_{2j}, \ldots, a_{mj})' \in \mathbb{R}^m$
- ith row vector $(a_{i1}, a_{i2}, \ldots, a_{in})' \in \mathbb{R}^n$

Notes

Basic operations

- Addition: $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$, then $\boldsymbol{A} - \boldsymbol{B} = (a_{ij} - b_{ij})$
- Scalar multiplication: $c \in \mathbb{R}$, $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, then $c\boldsymbol{A} = \boldsymbol{A}c = (ca_{ij})$
- Transposition: $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ with $\boldsymbol{A} = (a_{ij})$, then $\boldsymbol{A}' = (a_{ji})$. Note that $\boldsymbol{A}' \in \mathbb{R}^{n \times m}$

A key operation is the matrix-product. Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{n \times p}$ (compatible), then the matrix-product $\boldsymbol{P} \in \mathbb{R}^{m \times p}$ is given

$$(p_{ij}) = \langle \boldsymbol{A}_{i\cdot} \boldsymbol{B}_{\cdot j} \rangle = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Note that p_{ij} is the inner product between the i-th row of $\boldsymbol{A}$ and the j-th column of $\boldsymbol{B}$

⚠ in general $\boldsymbol{AB} \neq \boldsymbol{BA}$ (matrix multiplication is not commutative)

Suggestion: take any math book and do few matrix product calculations to warm up

Notes

Useful properties

- Associative for matrix-product: $\boldsymbol{A}(\boldsymbol{BC}) = (\boldsymbol{AB})\boldsymbol{C}$
- Distributive for matrix-product:
  - $\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{AB}+\boldsymbol{AC}$
  - $(\boldsymbol{B}+\boldsymbol{C})\boldsymbol{A} = \boldsymbol{BA}+\boldsymbol{CA}$
- For any $\delta \in \mathbb{R}$, $\delta(\boldsymbol{AB}) = (\delta\boldsymbol{A})\boldsymbol{B} = A(\delta\boldsymbol{B})$
- $(\boldsymbol{A}')' = \boldsymbol{A}$
- $(\boldsymbol{A}+\boldsymbol{B})' = \boldsymbol{A}'+\boldsymbol{B}'$
- $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$
- $(\boldsymbol{ABC})' = \boldsymbol{C}'\boldsymbol{B}'\boldsymbol{A}'$

Notes

Square matrices ($n = m$) are particularly important.

Diagonal matrix: is a matrix with all zeros outside the main diagonal

$$\boldsymbol{A} = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}$$

A particularly important diagonal matrix is the **identity matrix of order** n

$$\boldsymbol{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

For any $\boldsymbol{A}$ compatible with $\boldsymbol{I}_n$ we have that $\boldsymbol{I}_n\boldsymbol{A} = \boldsymbol{A}$ and $\boldsymbol{AI}_n = \boldsymbol{A}$

Notes

Symmetric matrices are square matrices with elements symmetrically placed around the diagonal, that is $a_{ij} = a_{ji}$ for all $i \neq j$:

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{12} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{nn} \end{pmatrix}$$

For symmetric matrices $\boldsymbol{A} = \boldsymbol{A}'$

Notes

An important real valued function in matrix computing is the **determinant**

$$\det : \{\text{Set of Square Matrices}\} \longrightarrow \mathbb{R}$$

denoted with either $\det(\boldsymbol{A})$ or $|\boldsymbol{A}|$.

There are various methods to compute the determinant, but it remains an horrible calculation to perform anyway. Fortunately we have computers these days!

However, $\det()$ is of huge interest because:

- it "encodes" several important properties of matrices
- it is essential to define the inverse of a matrix... in a minute you will see that the inverse closes the circle!

If $\det(\boldsymbol{A}) = 0$ we say that $\boldsymbol{A}$ is singular.

Notes

Useful properties

- $\det(\boldsymbol{A}) = \det(\boldsymbol{A}')$
- $\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B})$
- For $\delta \in \mathbb{R}$ and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\det(\delta\boldsymbol{A}) = \delta^n \det(\boldsymbol{A})$

Let $\boldsymbol{A}$ be a $(p \times p)$ square matrix, the following statements are equivalent:

- $\boldsymbol{A}$ is invertible
- $\det(\boldsymbol{A}) \neq 0$
- the columns/rows of $\boldsymbol{A}$ are linearly independent.
- the columns/rows of $\boldsymbol{A}$ span $\mathbb{R}^p$
- the columns/rows of $\boldsymbol{A}$ are a basis in $\mathbb{R}^p$.

Notes

A square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is called *invertible* or *non-singular* if there exists a (unique) matrix $\boldsymbol{B} \in \mathbb{R}^{n\times n}$ such that $\boldsymbol{AB} = \boldsymbol{BA} = \boldsymbol{I}_n$

In practice the inverse closes the circle of the matrix algebra, it allows to define the reciprocal element in our collection of objects!

In fact, when exists, the inverse of $\boldsymbol{A}$ is denoted by $\boldsymbol{A}^{-1}$.

An orthogonal matrix is a square real matrix whose columns and rows are orthogonal vectors with unit norm (orthonormal vectors). For orthogonal matrices

$$\boldsymbol{A}' = \boldsymbol{A}^{-1} \implies \boldsymbol{AA}' = \boldsymbol{A}'\boldsymbol{A} = \boldsymbol{I}$$

Notes

Useful properties

- $\det(\boldsymbol{A}^{-1}) = \frac{1}{\det(\boldsymbol{A})}$
- $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$
- $(\boldsymbol{ABC})^{-1} = \boldsymbol{C}^{-1}\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$
- $(\boldsymbol{A}')^{-1} = (\boldsymbol{A}^{-1})'$

Notes

Another real valued function of a square matrix is the *trace* = sum of diagonal elements. Let $\boldsymbol{A} = (a_{ij}) \in R^{n\times n}$, then

$$\mathrm{tr}(\boldsymbol{A}) = \sum_{i=1}^{n} a_{ii}$$

Useful properties

- $\mathrm{tr}(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A}')$
- $\mathrm{tr}(\boldsymbol{A}+\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{A}) + \mathrm{tr}(\boldsymbol{B})$
- $\mathrm{tr}(\boldsymbol{AB}) = \mathrm{tr}(\boldsymbol{BA})$
- $\mathrm{tr}(\boldsymbol{ABC}) = \mathrm{tr}(\boldsymbol{BCA}) = \mathrm{tr}(\boldsymbol{CAB})$ (cyclic property)
- For $\boldsymbol{x} \in \mathbb{R}^n$, then $\langle \boldsymbol{x}, \boldsymbol{x}\rangle = \|\boldsymbol{x}\|_2^2 = \boldsymbol{x}'\boldsymbol{x} = \mathrm{tr}(\boldsymbol{xx}')$

Notes

Rank of a matrix

The rank $\boldsymbol{A}$, denoted by $\mathrm{rank}(\boldsymbol{A})$, is the dimension of the vector space spanned by its columns/rows.

Therefore, the column/row rank of $\boldsymbol{A}$ is the number of linearly independent column/rows vectors of $\boldsymbol{A}$

It can be shown that column rank and row rank are always equal. Therefore $\mathrm{rank}\,\boldsymbol{A}$ conventionally denotes the column rank.

Notes

Assume $\boldsymbol{A}$ is a $n \times K$ matrix

- $\mathrm{rank}(\boldsymbol{A}) \leq \min(n, K)$
- if $\mathrm{rank}(\boldsymbol{A}) = \min(n, K)$ is said to be full-rank
- $\mathrm{rank}(\boldsymbol{A}) = \mathbf{0}$ if and only if $\boldsymbol{A}$ is the zero matrix
- if $\boldsymbol{A}$ is square matrix, then $\boldsymbol{A}$ is invertible if and only if $\boldsymbol{A}$ is full-rank.

Additionally, for any proper matrix $\boldsymbol{B}$

- $\mathrm{rank}(\boldsymbol{AB}) \leq \min\{\mathrm{rank}(\boldsymbol{A}), \mathrm{rank}(\boldsymbol{B})\}$
- $\mathrm{rank}(\boldsymbol{A}) + \mathrm{rank}(\boldsymbol{B}) \leq \mathrm{rank}(\boldsymbol{AB}) + K$
- $\mathrm{rank}(\boldsymbol{A} + \boldsymbol{B}) \leq \mathrm{rank}(\boldsymbol{A}) + \mathrm{rank}(\boldsymbol{B})$

Notes

Systems of linear equations

A general system of n linear equations with K unknowns $\{x_1, x_2, \ldots, x_K\}$ can be written as

$$\begin{cases} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1K}x_K & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2K}x_K & = & b_2 \\ & & & & & & & \vdots & \\ a_{n1}x_1 & + & a_{n2}x_2 & + & \cdots & + & a_{nK}x_K & = & b_n \end{cases}$$

A solution of the system (if it exists) is a a vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_K)'$ such that the equations hold jointly

There are three possibilities

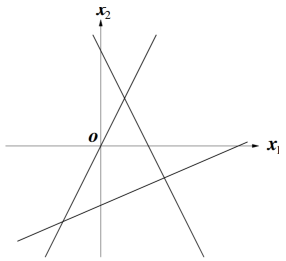
- the system does not have a solution
- the system has a unique solution
- the system has infinitely many solutions

Notes

The following system of 3 equations in 2 unknowns

$$\begin{cases} 4x_1 & + & 2x_2 & = & 6 \\ 4x_1 & - & 2x_2 & = & 0 \\ 2x_1 & - & 4x_2 & = & 8 \end{cases} \Longrightarrow \begin{cases} x_2 & = & -2x_1 + 3 \\ x_2 & = & 2x_1 \\ x_2 & = & \frac{1}{2}x_1 - 2 \end{cases}$$

Each of these is a straight line in $\mathbb{R}^2$. A solution will be a point where all three straight lines intersect. A solution does not exist:

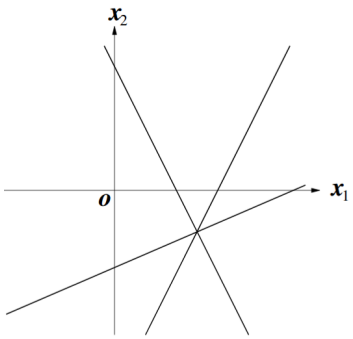


Notes

Now consider this system

$$\begin{cases} 6x_1 & + & 6x_2 & = & 9 \\ 6x_1 & - & 3x_2 & = & 15 \\ 3x_1 & - & 6x_2 & = & 12 \end{cases}$$

We have more equations than unknown, but there is a unique solution

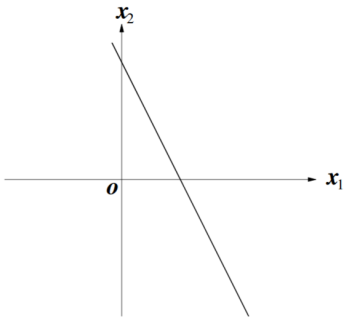


Notes

Now consider this system

$$\begin{cases} x_1 & + & 0.5x_2 & = & 1.5 \\ 2x_1 & + & x_2 & = & 3 \\ 3x_1 & + & 1.5x_2 & = & 4.5 \end{cases}$$

Again, more equations than unknown, but now all the equations define the same straight line therefore there are infinitely many solutions



Notes

Define the column vectors

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_K \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}, \qquad \boldsymbol{a}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{bmatrix}$$

An the coefficient matrix

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1K} \\ a_{21} & a_{22} & \cdots & a_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nK} \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_K \\ | & | & & | \end{bmatrix}$$

Notes

The augmented matrix

$$\tilde{\boldsymbol{A}} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1K} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2K} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \\ a_{n1} & a_{n2} & \cdots & a_{nK} & b_n \end{bmatrix} = \begin{bmatrix} | & | & & | & | \\ \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_K & \boldsymbol{b} \\ | & | & & | & | \end{bmatrix}$$

The system can be written in vector equation form

$$x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \ldots + x_K\boldsymbol{a}_K = \boldsymbol{b}$$

Or in matrix equation form

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$$

Notes

Conditions for the existence and uniqueness of solutions are given in the famous Rouché-Capelli theorem:

- a system of linear equations with K variables has a solution if and only if the $\mathrm{rank}(\boldsymbol{A}) = \mathrm{rank}(\tilde{\boldsymbol{A}})$
- if a solution exists and $\mathrm{rank}(\boldsymbol{A}) = K$, then the solution is unique
- if a solution exists and $\mathrm{rank}(\boldsymbol{A}) < K$, then there is an infinite number of solutions
- if a solution exists and $\mathrm{rank}(\boldsymbol{A}) < K$, the set of solutions forms a vector space of dimension $K - \mathrm{rank}(\boldsymbol{A})$

There are various algorithms to solve linear systems. Note that if $K = n$ and $\boldsymbol{A}^{-1}$ exists, the unique solution of the system can be easily computed as $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$. Why such solution exists?

Notes

Idempotent matrices and projections

A square matrix $\boldsymbol{A}$ is idempotent iff $\boldsymbol{A}^2 = \boldsymbol{A}\boldsymbol{A} = \boldsymbol{A}$

- The only full rank, symmetric idempotent matrix is the identity $\boldsymbol{I}$
- If $\boldsymbol{A} \neq \boldsymbol{I}$ is symmetric idempotent $\implies \det(\boldsymbol{A}) = 0$
- If $\boldsymbol{A}$ is symmetric idempotent, then $\mathrm{rank}(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A})$

A projection is a linear transformation from a vector space to itself. Suppose $\boldsymbol{x} \in \mathbb{R}^K$ then a projection looks like

$$\boldsymbol{y} = \boldsymbol{P}\boldsymbol{x}$$

this transforms $\boldsymbol{x}$ in another element of the same space.

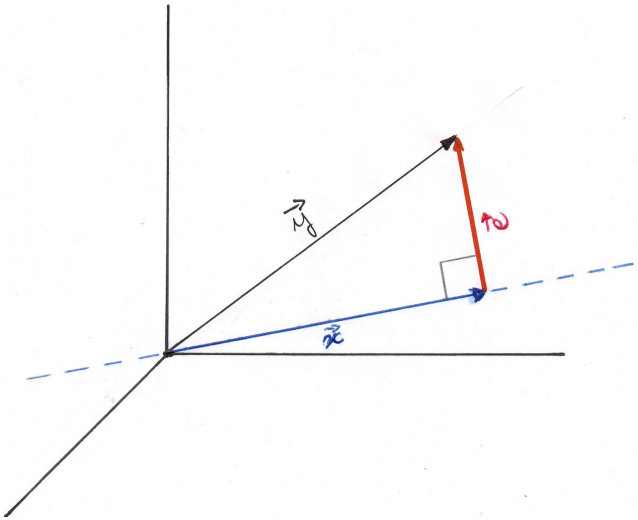
The projection has to fulfill the "primitive notion" that "we dont't change anything if we re-project the same object twice", so we want that

$$\boldsymbol{P}\boldsymbol{y} = \boldsymbol{y} \quad \implies \quad \boldsymbol{P}(\boldsymbol{P}\boldsymbol{x}) = \boldsymbol{P}^2\boldsymbol{x} = \boldsymbol{P}\boldsymbol{x}$$

$\implies$ that a projection matrix needs to be idempotent

Notes

Orthogonal projections



Notes

Quadratic forms

Let $\boldsymbol{x} \in \mathbb{R}^p$ and $\boldsymbol{A} \in \mathbb{R}^{p \times p}$, $\boldsymbol{A}$ is symmetric, the associated quadratic form is a function $q_{\boldsymbol{A}} : \mathbb{R}^p \to \mathbb{R}$

$$q_{\boldsymbol{A}}(\boldsymbol{x}) = \sum_{i=1}^{p} \sum_{j=1}^{p} a_{ij} x_i x_j = \boldsymbol{x}' \boldsymbol{A} \boldsymbol{x}$$

- A real symmetric matrix $\boldsymbol{A}$ is positive definite (write PD or $\boldsymbol{A} \succ 0$) if for all $\boldsymbol{x} \neq \boldsymbol{0}$ the quadratic form $\boldsymbol{x}' \boldsymbol{A} \boldsymbol{x} > 0$
- $\boldsymbol{A}$ is positive semi-definite (write PSD or $\boldsymbol{A} \succeq 0$) if for all $\boldsymbol{x} \neq \boldsymbol{0}$ the quadratic form $\boldsymbol{x}' \boldsymbol{A} \boldsymbol{x} \geq 0$
- If $\boldsymbol{A} \succ 0$ then $\det(\boldsymbol{A}) > 0$, therefore PD matrix are non-singular

Loewner ordering. $\boldsymbol{A} \succeq \boldsymbol{B}$ means that $\boldsymbol{A} - \boldsymbol{B} \succeq 0$, that is $\boldsymbol{A} - \boldsymbol{B}$ is PSD, or $\boldsymbol{x}' \boldsymbol{A} \boldsymbol{x} \geq \boldsymbol{x}' \boldsymbol{B} \boldsymbol{x}$ for all $\boldsymbol{x} \neq \boldsymbol{0}$. This formalizes the concept that "$\boldsymbol{A}$ is overall bigger than $\boldsymbol{B}$"

Notes

Multivariate probability

In this course we consider vector valued random variables (rv) , take a rv $X = (X_1, X_2, \ldots, X_p)'$ with values in $\mathbb{R}^p$.

The probability law of a (vector) rv is represented by its multivariate distribution function, for $\boldsymbol{t} \in \mathbb{R}^p$

$$F(\boldsymbol{t}) = \Pr\{X_1 \leq t_1, X_2 \leq t_2, \ldots, X_p \leq t_p\}$$

In most cases F is smooth enough that has a *derivative* f called the density function at any $\boldsymbol{x} \in \mathbb{R}^p$

$$f(\boldsymbol{x}) = f(x_1, x_2, \ldots, x_p) = \left.\frac{\partial F(t_1, t_2, \ldots, t_p)}{\partial t_1 \partial t_2, \cdots \partial t_p}\right|_{\boldsymbol{t}=\boldsymbol{x}}$$

These are *absolutely continuous* rv, for which

$$F(\boldsymbol{t}) = \int_{-\infty}^{t_1} \int_{-\infty}^{t_2} \cdots \int_{-\infty}^{t_p} f(x_1, x_2, \ldots, x_p) dx_1 dx_2 \cdots dx_p$$

Notes

Multivariate moments

First order moments, that is, unconditional expectation can be easily extended from the univariate case. For a rv $X = (X_1, X_2, \ldots, X_p)$ in $\mathbb{R}^p$

$$\mathrm{E}[X] = (\mathrm{E}[X_1], \mathrm{E}[X_2], \ldots, \mathrm{E}[X_p])'$$

$\mathrm{E}[X]$ is a centrality/location measure.

Definition of second order moments is less obvious. There are several possibilities. We work with $\mathrm{E}[XX']$, but the argument is a $(p \times p)$-matrix, so here we mean expectation element-wise. For example take $X \in \mathbb{R}^2$

$$\mathrm{E}[XX'] = \mathrm{E}\begin{bmatrix} X_1^2 & X_1X_2 \\ X_2X_1 & X_2^2 \end{bmatrix} = \begin{pmatrix} \mathrm{E}[X_1^2] & \mathrm{E}[X_1X_2] \\ \mathrm{E}[X_2X_1] & \mathrm{E}[X_2^2] \end{pmatrix}$$

Exercise: let $X \in \mathbb{R}^p$ write down the second moment of X (as above)

Notes

What about variance? Consider X in $\mathbb{R}^p$ with expectation $\boldsymbol{\mu} \in \mathbb{R}^p$. Then for all $i, j = 1, 2, \ldots, p$

$$\sigma_{i,j} = \mathrm{E}[(X_i - \mu_i)(X_j - \mu_j)] = \mathrm{E}[X_i X_j] - \mathrm{E}[X_i]\,\mathrm{E}[X_j]$$

The variance of X is summarized in the $(p \times p)$ variance-covariance matrix

$$\mathrm{Var}(X) = \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,p} \\ \sigma_{2,1} & \sigma_2^2 & \cdots & \sigma_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p,1} & \sigma_{p,2} & \cdots & \sigma_p^2 \end{pmatrix}$$

Usually we say "covariance matrix" or simply "variance of X"

The covariance matrix is connected to its second order moment:

$$\boldsymbol{\Sigma} = \mathrm{E}[(X - \boldsymbol{\mu})(X - \boldsymbol{\mu})'] = \mathrm{E}[XX'] - \boldsymbol{\mu}\boldsymbol{\mu}'$$

it describes a lot of things about the linear dependence between the margins

Notes

$\det(\boldsymbol{\Sigma})$ is called "generalized variance".

- large $\det(\boldsymbol{\Sigma})$: large overall dispersion around the mean vector.
- small $\det(\boldsymbol{\Sigma})$: the distribution of X is extremely concentrated around the mean vector.

Singular covariance, that is $\det(\boldsymbol{\Sigma}) = 0$

- we can find a direction in the Euclidean space where there is absence of variability. For example a marginal X_i is degenerate.
- some of the components of X are collinear

Usually we want to get rid of singular cases, and we always look to rv with a nice PD covariance.

Notes

Correlation matrix

The covariance matrix contains information about the correlation structure. Given $\boldsymbol{\Sigma}$, let

$$\boldsymbol{Q} = (\text{diag}(\boldsymbol{\Sigma}))^{-\frac{1}{2}} = \begin{pmatrix} 1/\sigma_1 & 0 & \cdots & 0 \\ 0 & 1/\sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_p \end{pmatrix}$$

The correlation matrix is simply given by

$$\text{Cor}(X) = \boldsymbol{R} = \boldsymbol{Q}\boldsymbol{\Sigma}\boldsymbol{Q}$$

Notes

Now check (wlog you can take $X \in \mathbb{R}^2$) that

$$\boldsymbol{R}_{(i,j)} = \rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j}$$

In general the correlation matrix of a rv $X \in \mathbb{R}^p$ is a $(p \times p)$ symmetric matrix

$$\text{Cor}(X) = \boldsymbol{R} = \begin{pmatrix} 1 & \rho_{1,2} & \cdots & \rho_{1,p} \\ \rho_{2,1} & 1 & \cdots & \rho_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p,1} & \rho_{p,2} & \cdots & 1 \end{pmatrix}$$

Notes

Sampling estimators of mean and covariance

Let $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ be n sampling replicas of the rv X defined in $\mathbb{R}^p$.

Sampling estimator are simply obtained by replacing the distribution of X (called population distribution) with the empirical distribution. The empirical distribution is a uniform (discrete) distribution that attaches a probability of $1/n$ to each sample point.

- sample mean: $\overline{\boldsymbol{x}}_n = \frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i$
- sample cov (biased):

$$\begin{aligned}\boldsymbol{S}_n =& \frac{1}{n}\sum_{i=1}^n (\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)' \\ =& \left(\frac{1}{n}\sum_{i=1}^n \boldsymbol{x}_i \boldsymbol{x}_i'\right) - \overline{\boldsymbol{x}}_n \overline{\boldsymbol{x}}_n{}'\end{aligned}$$

- sample cov (unbiased): $\boldsymbol{S}_n = \frac{1}{n-1}\sum_{i=1}^n (\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)(\boldsymbol{x}_i - \overline{\boldsymbol{x}}_n)'$

Notes

Orthogonality: probabilistic viewpoint

Consider the set

$$\mathbb{H} = \{X : X \text{ is real-valued random variable such that } \mathrm{E}[X^2] < +\infty\}$$

This is set can be framed as vector space with an inner product! Therefore where one can define notions of *length/size*, *distance angle*... and orthogonality as we do for the Euclidean space.

A simple way to define an inner product on $\mathbb{H}$ is

$$\langle X, Y\rangle = \mathrm{E}[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])] = \mathrm{Cov}(X, Y)$$

We say X and Y are orthogonal if $\mathrm{Cov}(X, Y) = 0$

⚠ For random variables we use orthogonality as equivalent to linear independence. This is confusing if we think about the geometric definition.

Notes

The confusion needs to be fixed:

- $\mathrm{Cov}(X, Y) = 0 \implies X$ and Y are two orthogonal members of $\mathbb{H}$. It does not exist conditional mean dependence between X, Y
- $|\mathrm{Cov}(X, Y)| > 0 \implies X$ and Y are non-orthogonal. In this case there may be a *linear conditional mean* representation of the dependence. That's why we say that there is "*linear dependence*"

The general abstract notion of *orthogonality* is also essential in linear statistical models. This will be understood as course progresses

Notes

Linear maps between random variables

Linear transformations are common in Econometrics. Take a rv $X \in \mathbb{R}^p$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. Consider $\boldsymbol{t} \in \mathbb{R}^p$ and a $(p \times p)$ matrix $\boldsymbol{A}$.

Consider another rv $Y \in \mathbb{R}^p$ obtained as a linear map

$$Y = \boldsymbol{t} + \boldsymbol{A}X$$

- $\mathrm{E}[Y] = \boldsymbol{t} + \boldsymbol{A}\boldsymbol{\mu}$
- $\mathrm{Var}[Y] = \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}'$

Notes

Multivariate Normal Distribution (MVN)

The probability law is governed by first and second order moments. $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ has density

$$\phi(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{\det(\boldsymbol{\Sigma})^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}$$

Where $\mathrm{E}[X] = \boldsymbol{\mu}$ and $\mathrm{Cov}[X] = \boldsymbol{\Sigma}$.

For $\boldsymbol{\mu} = \boldsymbol{0}$ and $\boldsymbol{\Sigma} = \boldsymbol{I}_p$ we obtain the standard (spherical) MVN.

A MNV produces elliptically symmetric scatters.

Notes

The MVN is extremely important in the "linear world" because of the following properties. Assume $X \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

- Any linear combination of X is again MVN. If $Y = \boldsymbol{t} + \boldsymbol{A}X$ then $Y \sim \mathcal{N}(\boldsymbol{t} + \boldsymbol{A}\boldsymbol{\mu}, \boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}')$
- Any subset of marginals of X have a joint MVN distribution
- Uncorrelation $\Longleftrightarrow$ independence
- Conditional distributions are MVN

Notes

Quadratic forms of rv

An important quadratic form is the "squared Mahalanobis distance" (SMD)

$$(\boldsymbol{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})$$

this gives the squared "weighted" distance of $\boldsymbol{x}$ from $\boldsymbol{\mu}$ by taking into account the correlation structure.

Note that in the spherical case that $\boldsymbol{\Sigma}=\boldsymbol{I}_p$, the SMD becomes

$$(\boldsymbol{x}-\boldsymbol{\mu})'(\boldsymbol{x}-\boldsymbol{\mu})=\|\boldsymbol{x}-\boldsymbol{\mu}\|_2^2=d_2(\boldsymbol{x},\boldsymbol{\mu})^2$$

In the SMD the $\boldsymbol{\Sigma}^{-1}$ is used to "rotate" the axis so that the scatter becomes spherical, and then the usual Euclidean norm is used to measure distances between points.

Notes

Quadratic forms of rv with MNV distribution occur all the time in Econometrics. The following results will be used for the rest of the course:

If $X\sim\mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ then $(\boldsymbol{x}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\sim\chi^2(p)$

A special case is when If $Z\sim\mathcal{N}(\boldsymbol{0},\boldsymbol{I}_p)$, then $Z'Z\sim\chi^2(p)$

The more general result: if $Z\sim\mathcal{N}(\boldsymbol{0},\boldsymbol{I}_p)$, $\boldsymbol{A}$ is idempotent (that is $\boldsymbol{A}\boldsymbol{A}=\boldsymbol{A}$), rank$(\boldsymbol{A})=m$, then $Z'\boldsymbol{A}Z\sim\chi^2(m)$

Notes

Conditional expectation

Suppose that X is a continuous/discrete rv, and Y is and a discrete rv. We all know what $\mathrm{E}[X \,|\, Y = y]$ means and we know how to compute it

For instance, assume that X is continuous and you know $f_X(x \,|\, Y = y)$

$$\mathrm{E}[X \,|\, Y = y] = \int x \, f_X(x \,|\, Y = y) dx$$

Suppose now that Y is continuous, and consider $\mathrm{E}[X \,|\, Y]$. Have you ever thought about what this really means?

If not, you should seriously think about it, because 99% of econometric models are specified in terms of conditional expectations like that

Notes

In elementary statistics you learned $\Pr\{A \,|\, B\}$ where A and B are events

And in fact, we only know how to condition on events, not on a random variable (stochastic conditioning)

It turns out that stochastic conditioning is the same as the usual conditioning

In order to better understand the latter, we need to better understand how we formalize the information generated from an experiment

Notes

Intuitive notion of information

Consider a probability space $(\Omega, \mathcal{F}, P)$

- Ω is the corresponding sample/outcome space
- $\mathcal{F}$ is the set of all possible events for which you are able to compute probabilities
- P assigns probabilities to the events

$\mathcal{F}$ is the "*biggest information set*" you can manage with P, it represents "*knowing everything*"

Any $\mathcal{H} \subseteq \mathcal{F}$ is a "*smaller information set*", it represents "*knowing something*"

$\mathcal{B} = \{\emptyset, \Omega\}$ is the "*smallest information set*" and it is contained in any $\mathcal{H} \subseteq \mathcal{F}$, it represents "*knowing nothing*"

Notes

Intuitive notion of conditional expectation

Consider a rv X, you are asked to compute its (unconditional) expectation $\mathrm{E}[X]$. Assuming you know the probability law of X, you know how to do it.

Now suppose that events in $\mathcal{H}$ occurred. You don't want to waste this information, so you immediately update the probability law of X, with the posterior probability law of X.

Now you are asked to recompute the expectation of X. What do you do? Since you know how to use information efficiently, you redo your calculations replacing the original probability of X with the previous posterior law.

Doing this, your are just computing the conditional expectation

$$\mathrm{E}[X \mid \mathcal{H}]$$

Notes

Stochastic conditioning

Let X and Y be rv. Define the information set generated when we perform the experiment Y as

$$\sigma(Y) \subseteq \mathcal{F}$$

This information can be seen as

- $\sigma(Y) =$ information contained in Y
- $\sigma(Y) =$ what we know when we see one of the many results of Y
- for serious math-nerd $\sigma(Y) = \sigma-$*algebra* generated by Y

We write $\mathrm{E}[X \mid Y]$, but this is (a terribly bad) short notation for $\mathrm{E}[X \mid \sigma(Y)]$. Summarizing

- stochastic conditioning is not that different from the usual concept
- $\mathrm{E}[X \mid Y] = g(Y)$ is a rv itself: each random experimental result produced by Y will produce a different value for $\mathrm{E}[X \mid Y]$

Notes

Note that

$$\textit{unconditional expectation} = \textit{knowing nothing}$$

where

$$\textit{knowing nothing} = \mathcal{B} = \{\emptyset, \Omega\}$$

In fact, $\mathrm{E}[X]$ is also short notation for

$$\mathrm{E}[X] = \mathrm{E}[X \mid \mathcal{B}] = \mathrm{E}[X \mid \{\emptyset, \Omega\}]$$

It is also now intuitive why $\mathrm{E}[Y \mid \mathcal{F}] = Y$

Notes

Law of total expectation (aka Tower rule)

If X and Y are rv, we know that $\mathrm{E}[X \,|\, Y] = g(Y)$ is a rv. Therefore it make sense to compute expectations of $\mathrm{E}[X \,|\, Y]$

Math statement: consider two "smaller" information sets: $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{F}$, then

$$\mathrm{E}[\mathrm{E}[X \,|\, \mathcal{H}_2] \,|\, \mathcal{H}_1] = \quad \mathrm{E}[X \,|\, \mathcal{H}_1] \quad = \mathrm{E}[\mathrm{E}[X \,|\, \mathcal{H}_1] \,|\, \mathcal{H}_2]$$

Practical interpretation: *in sequential conditioning the smallest information set always wins!*

As a special case we obtain the "*Law of iterated expectation*"

Notes

Law of iterated expectation

Statement: consider two rv X and Y, then

$$\mathrm{E}[\mathrm{E}[X \,|\, Y]] = \mathrm{E}[X]$$

To see why this is a consequence of the previous result, define

- $\mathcal{H}_2 = \sigma(Y) \subseteq \mathcal{F}$
- $\mathcal{H}_1 = \mathcal{B} = \{\emptyset, \Omega\}$, and by definition $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{F}$

The *Law of total expectation* implies

$$\mathrm{E}[\ \mathrm{E}[X \,|\, \mathcal{H}_2] \,|\, \mathcal{H}_1\ ] = \mathrm{E}[X \,|\, \mathcal{H}_1]$$

That is

$$\mathrm{E}[\ \mathrm{E}[X \,|\, \sigma(Y)] \,|\, \mathcal{B}\ ] = \mathrm{E}[X \,|\, \mathcal{B}]$$

Using the "*short notation*" for $\sigma(Y)$, and omitting $\mathcal{B}$ as usual

$$\mathrm{E}[\ \mathrm{E}[X \,|\, Y]\ ] = \mathrm{E}[X]$$

Notes

Pull out what's known

Another useful property of conditional expectation is the so called "Pull out what's known" (sometimes called "linearity of conditional expectations"):

$$\mathrm{E}[XY \mid Y] = Y\,\mathrm{E}[X \mid Y]$$

Can you formulate any intuitive explanation of this result?

Notes

Notes